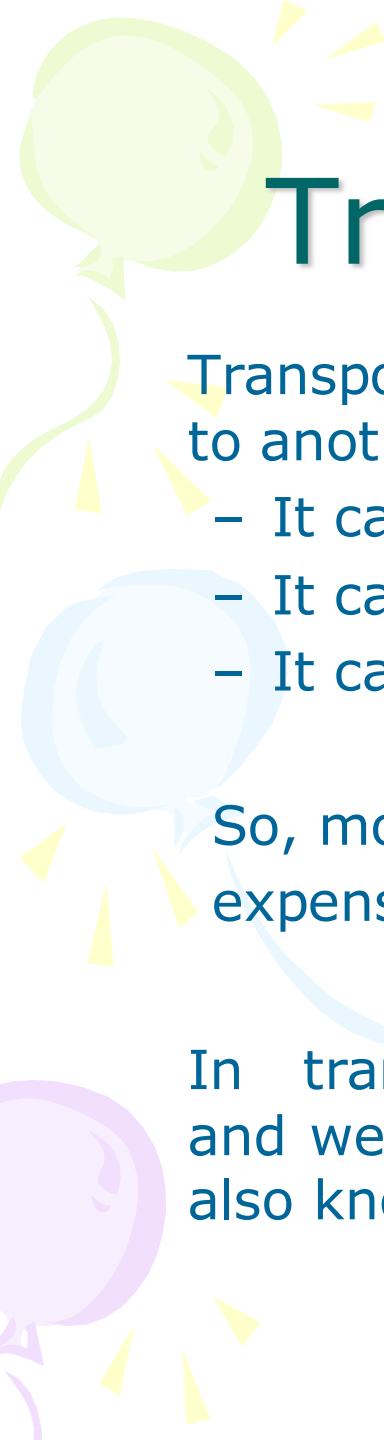


The background features several large, flowing, abstract shapes in light green, light blue, and light purple. Interspersed among these are numerous small, yellow, triangular starburst or confetti-like shapes, some of which are clustered together.

# **TRANSPORTATION PROBLEM**

**Finding Initial Basic Feasible Solution**



# Transportation Problem

Transportation means movement of goods from one place to another place.

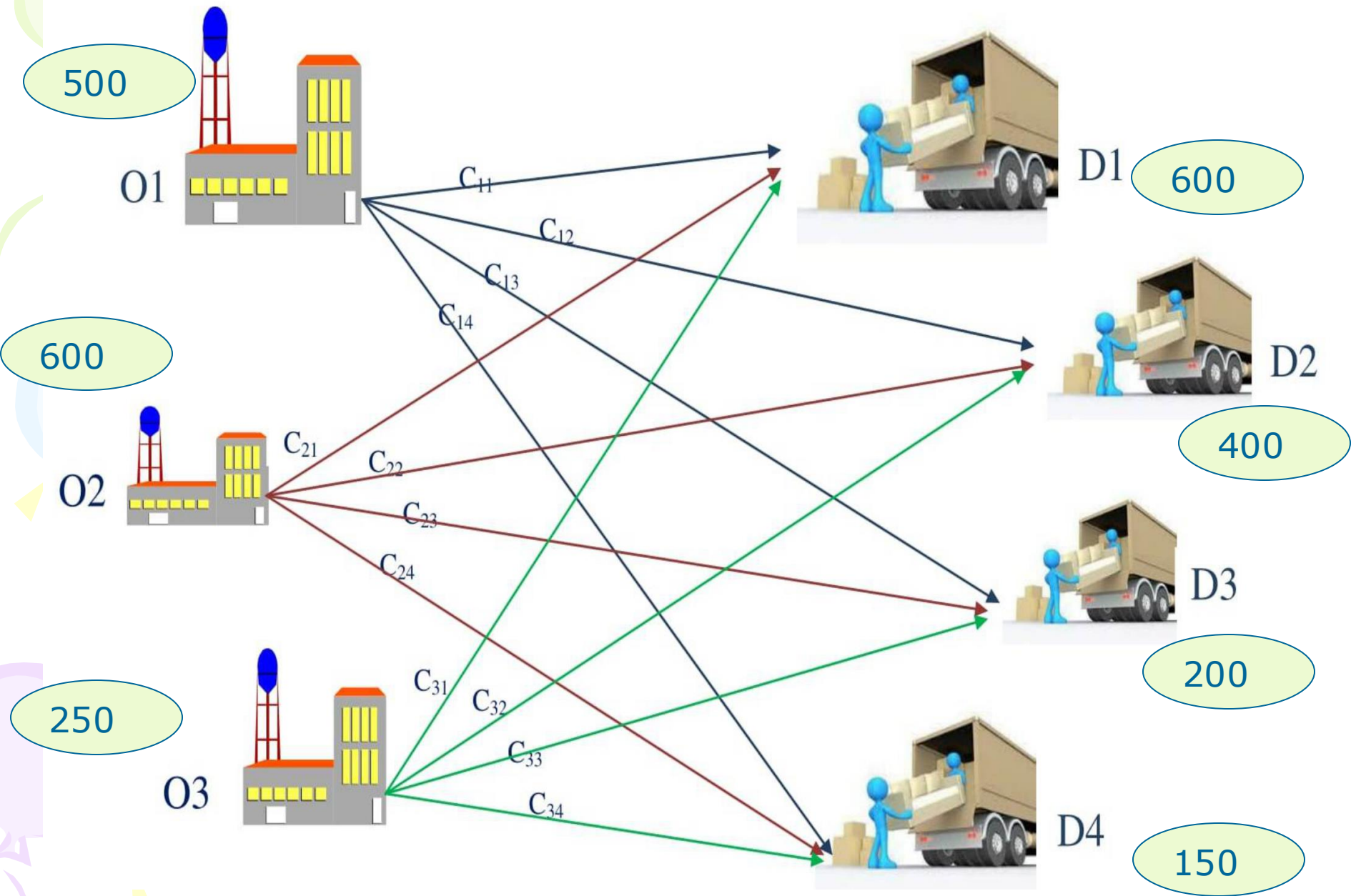
- It can be from one city to the next city.
- It can be from one place to the next place.
- It can be from one country to another country.

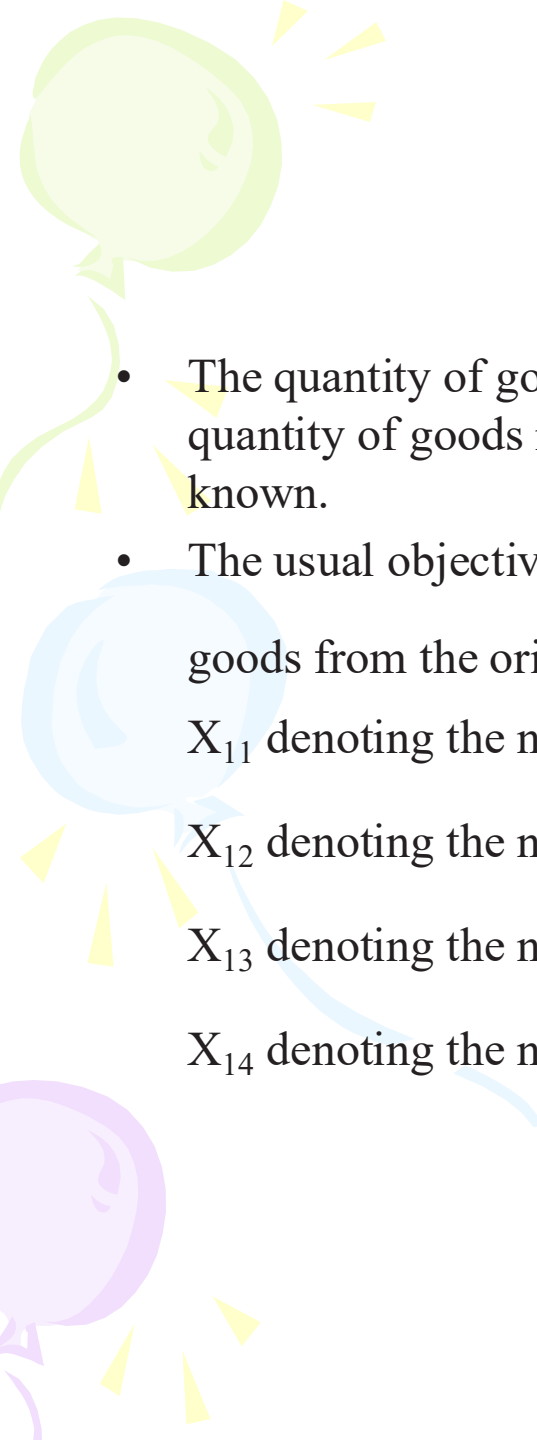
So, movement is going to incur some costs or some expenses and we always try to minimize this expenses.

In transportation of goods its going to incur some costs and we are going to minimize that costs that is why this is also known as **minimization type of problem.**

m-supply points

n-demand points





# Formulation

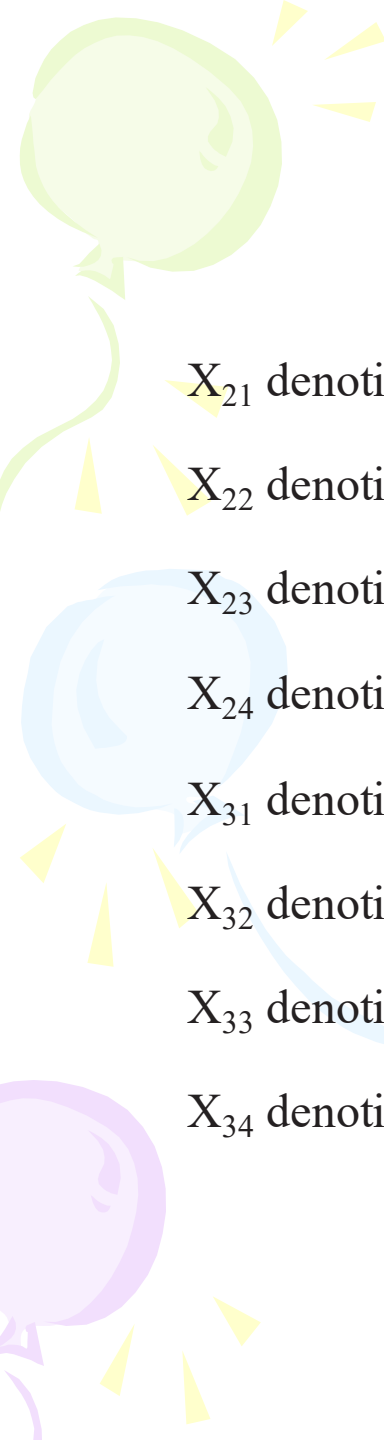
- The quantity of goods available at each supply location (origin) is limited, and the quantity of goods needed at each of several demand locations (destinations) is known.
- The usual objective in a transportation problem is to minimize the cost of shipping goods from the origins to the destinations.

$X_{11}$  denoting the number of units shipped from Origin 1 to Destination 1

$X_{12}$  denoting the number of units shipped from Origin 1 to Destination 2

$X_{13}$  denoting the number of units shipped from Origin 1 to Destination 3

$X_{14}$  denoting the number of units shipped from Origin 1 to Destination 1



# Formulation

$X_{21}$  denoting the number of units shipped from Origin 2 to Destination 1

$X_{22}$  denoting the number of units shipped from Origin 2 to Destination 2

$X_{23}$  denoting the number of units shipped from Origin 2 to Destination 3

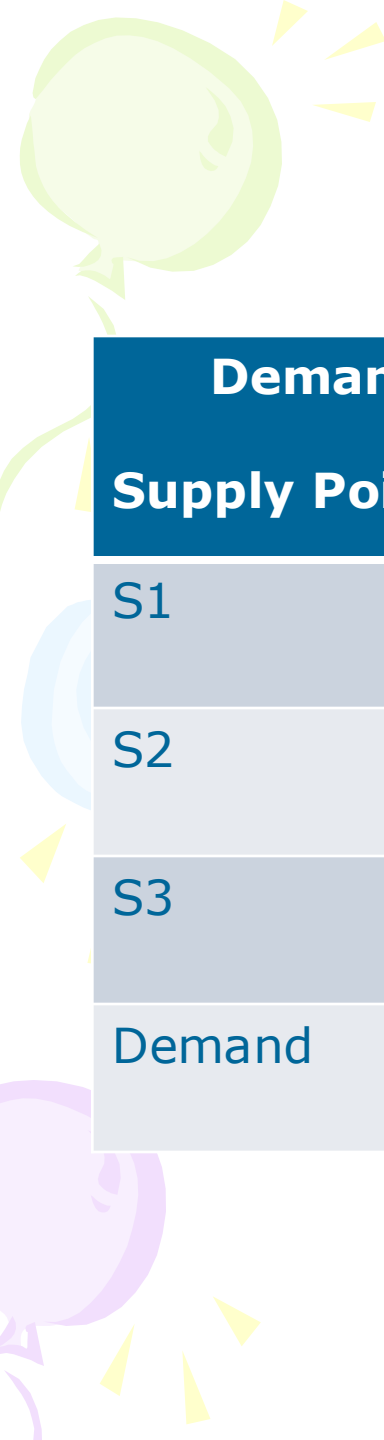
$X_{24}$  denoting the number of units shipped from Origin 2 to Destination 4

$X_{31}$  denoting the number of units shipped from Origin 3 to Destination 1

$X_{32}$  denoting the number of units shipped from Origin 3 to Destination 2

$X_{33}$  denoting the number of units shipped from Origin 3 to Destination 3

$X_{34}$  denoting the number of units shipped from Origin 3 to Destination 4



| Demand points<br>Supply Points | D1         | D2         | D3         | D4         | Supply         |
|--------------------------------|------------|------------|------------|------------|----------------|
| S1                             | C11<br>X11 | C12<br>X12 | C13<br>X13 | C14<br>X14 | 500            |
| S2                             | C21<br>X21 | C22<br>X22 | C23<br>X23 | C24<br>X24 | 600            |
| S3                             | C31<br>X31 | C32<br>X32 | C33<br>X33 | C34<br>X34 | 250            |
| Demand                         | 600        | 400        | 200        | 150        | Total=<br>1350 |



# Formulation

$$\text{MIN. } Z = C_{11}X_{11} + C_{12}X_{12} + C_{13}X_{13} + C_{14}X_{14} + C_{21}X_{21} + C_{22}X_{22} + C_{23}X_{23} + C_{24}X_{24} + C_{31}X_{31} + C_{32}X_{32} \\ + C_{33}X_{33} + C_{34}X_{34}$$

$$X_{11} + X_{12} + X_{13} + X_{14} = 500$$

$$X_{21} + X_{22} + X_{23} + X_{24} = 600$$

$$X_{31} + X_{32} + X_{33} + X_{34} = 250$$

$$X_{11} + X_{21} + X_{31} = 600$$

$$X_{12} + X_{22} + X_{32} = 400$$

$$X_{13} + X_{23} + X_{33} = 200$$

$$X_{14} + X_{24} + X_{34} = 150$$


# Details of Transportation Problem

Sources

|                  | Destinations |                |                | Supply |
|------------------|--------------|----------------|----------------|--------|
|                  | Civil Mall   | Kathmandu Mall | People's Plaza |        |
| <u>Dhulikhel</u> | 2            | 3              | 4              | 10     |
| <u>Hetauda</u>   | 11           | 10             | 2              | 20     |
| <u>Bharatpur</u> | 4            | 5              | 6              | 30     |
| Demand           | 15           | 30             | 15             |        |

maximum supply which means that Dhulikhel based manufacturing company can supply maximum 10 units of shoes.

Maximum Demand which means Civil Mall has maximum demand of 15 units and Kathmandu Mall has maximum demand of 30 units.



# Details of Transportation Problem

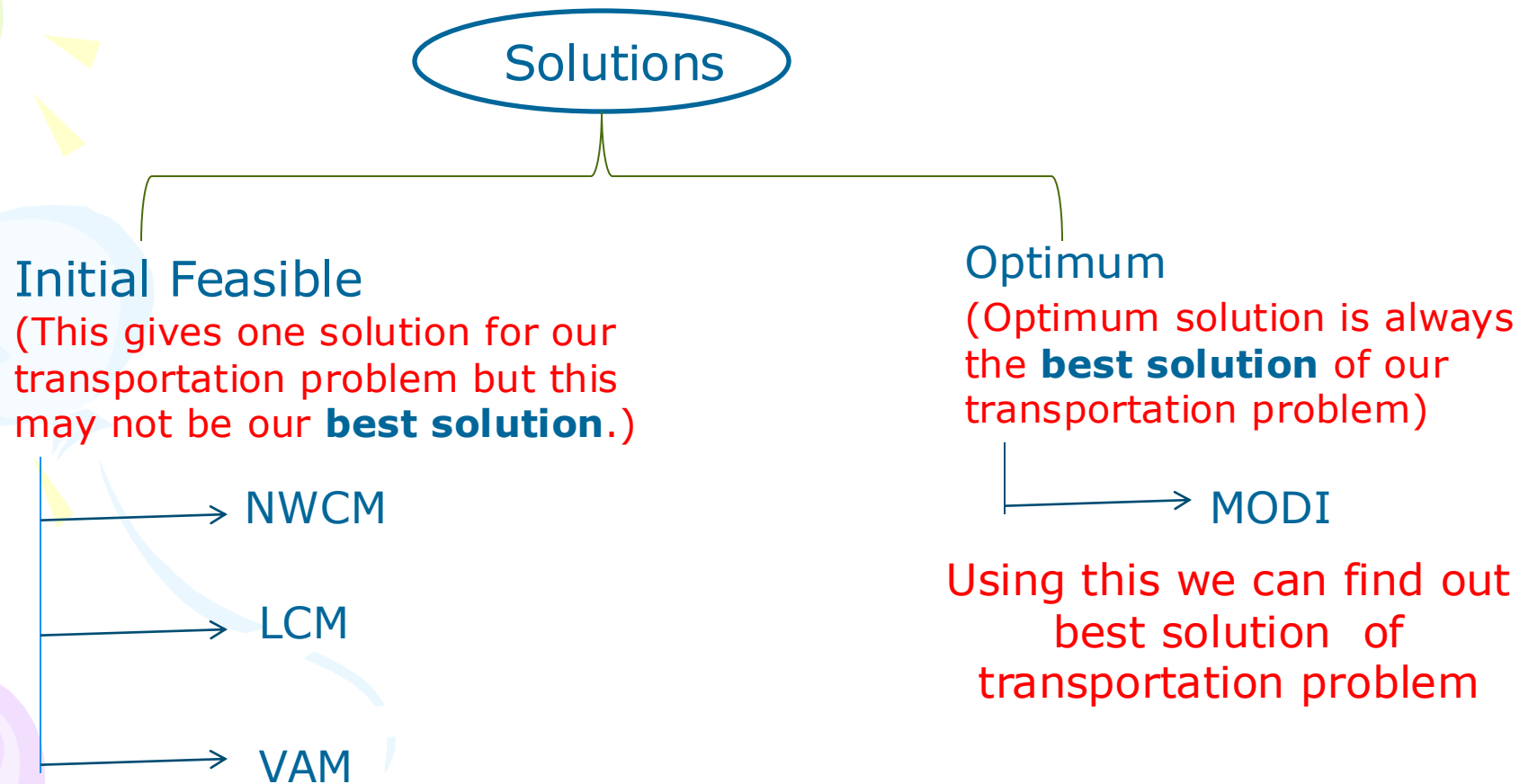
|                  | Civil Mall | Kathmandu Mall | People's Plaza | Supply  |
|------------------|------------|----------------|----------------|---------|
| <u>Dhulikhel</u> | 2          | 3              | 4              | 10      |
| <u>Hetauda</u>   | 11         | 10             | 2              | 20      |
| <u>Bharatpur</u> | 4          | 5              | 6              | 30      |
| Demand           | 15         | 30             | 15             | 60 / 60 |

When the total supply is equal to total demands than this is Balanced type of matrix / **Balanced transportation problem.**

While solving transportation problem if we have **Unbalanced matrix** then we should balance it and then solve it.

Aim of transportation problem is to find out minimum costs in transporting shoes from Different city-based manufacturing company to different malls in Kathmandu city.

# How to solve Transportation Problem ?



# How to solve Transportation Problem ?

- ▶ We cannot directly find out optimum solution, first step is to find out **Initial Basic Feasible Solution** and depending upon that solution we can infer if its optimum solution exists or not and then we may find optimal solution.

## Initial Feasible :

NWCM (Northwest – Corner Method)

- ▶ LCM (Least – Cost Method)

VAM (Vogel Approximation Method)

Condition for feasibility; no. of allocations =  $m+n-1$ ; where  $m$ = no. of rows and  $n$ = no. of columns

## Optimum :

MODI (Modified Distribution Method)

# Northwest – Corner Method



[dreamstime.com](http://dreamstime.com)

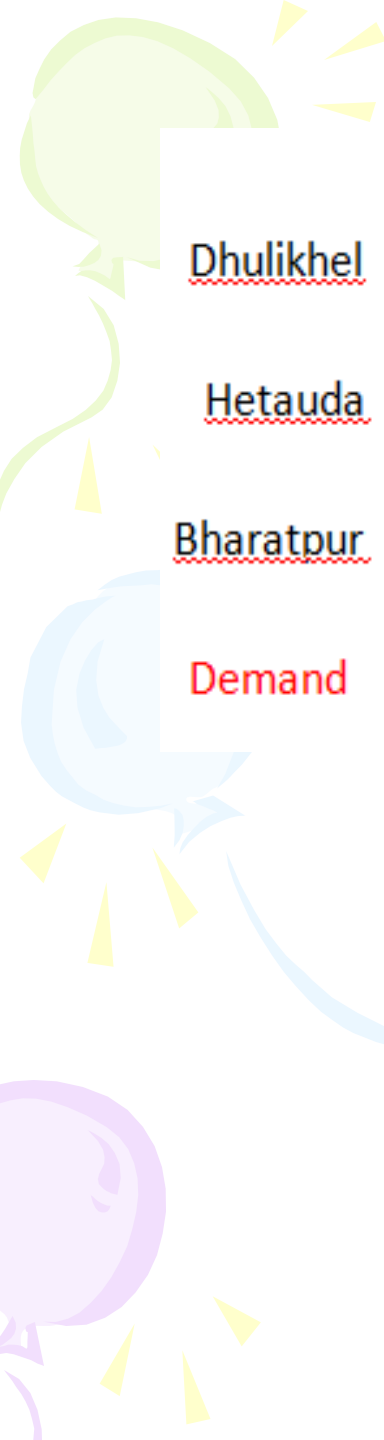
# North-West Corner Method

**Step1:** Select the upper left (north-west) cell of the transportation matrix and allocate the possible value to  $X_{11}$  which is equal to  $\min(a_1, b_1)$ .

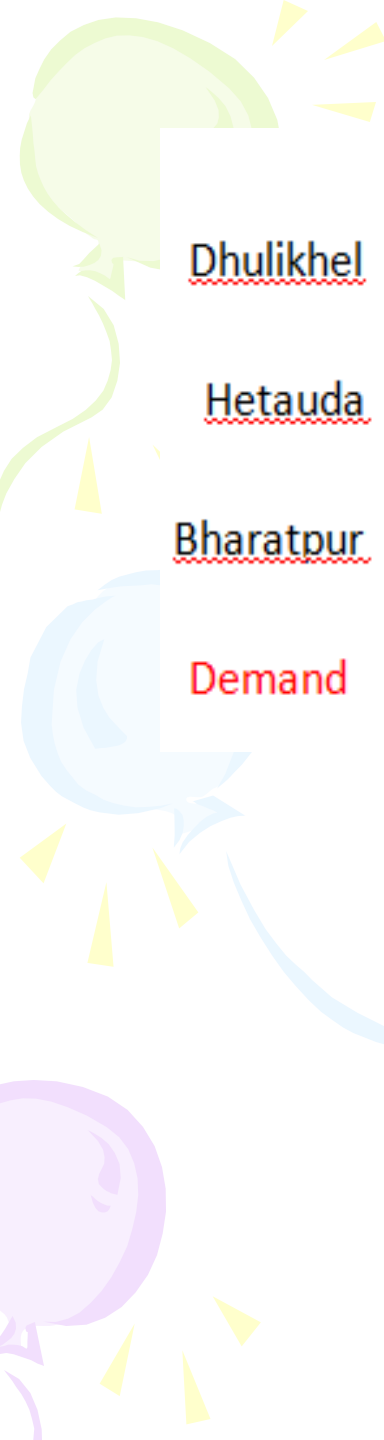
**Step2:**

- If allocation made is equal to the supply available at the first source ( $a_1$  in first row), then move vertically down to the cell (2,1).
- If allocation made is equal to demand of the first destination ( $b_1$  in first column), then move horizontally to the cell (1,2).
- If  $a_1 = b_1$ , then allocate  $X_{11} = a_1$  or  $b_1$  and move to cell (2,2).

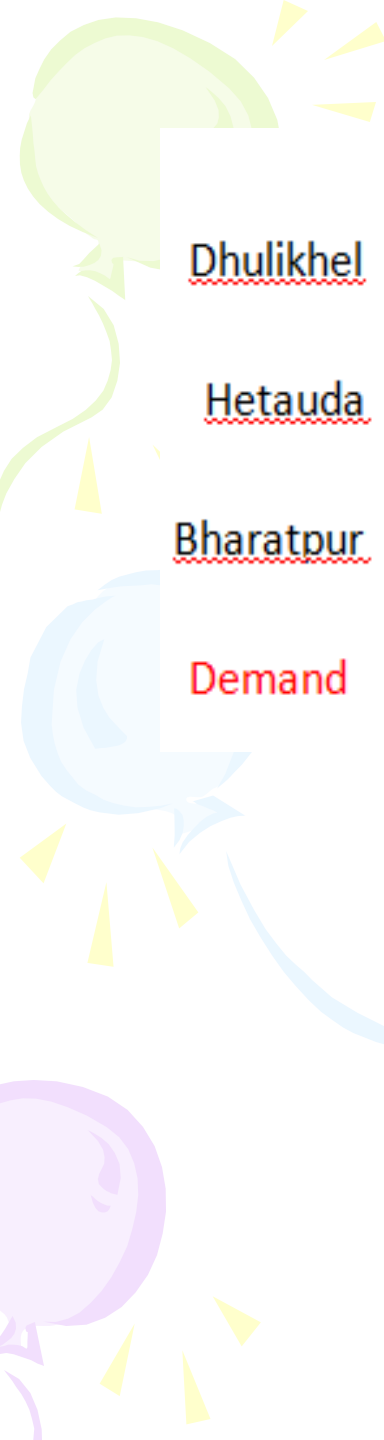
**Step3:** Continue the process until an allocation is made in the south-east corner cell of the transportation table.



|                  | Civil Mall | Kathmandu Mall | People's Plaza | Supply  |
|------------------|------------|----------------|----------------|---------|
| <u>Dhulikhel</u> | 2          | 3              | 4              | 10      |
| <u>Hetauda</u>   | 11         | 10             | 2              | 20      |
| <u>Bharatpur</u> | 4          | 5              | 6              | 30      |
| Demand           | 15         | 30             | 15             | 60 / 60 |

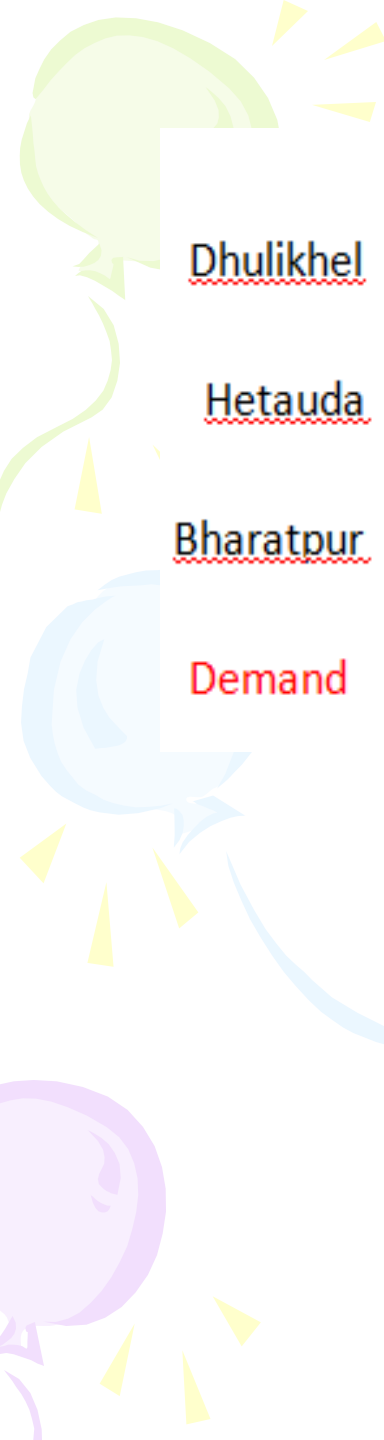


|                  | Civil Mall | Kathmandu Mall | People's Plaza | Supply  |
|------------------|------------|----------------|----------------|---------|
| <u>Dhulikhel</u> | 2          | 3              | 4              | 10      |
| <u>Hetauda</u>   | 11         | 10             | 2              | 20      |
| <u>Bharatpur</u> | 4          | 5              | 6              | 30      |
| Demand           | 15         | 30             | 15             | 60 / 60 |

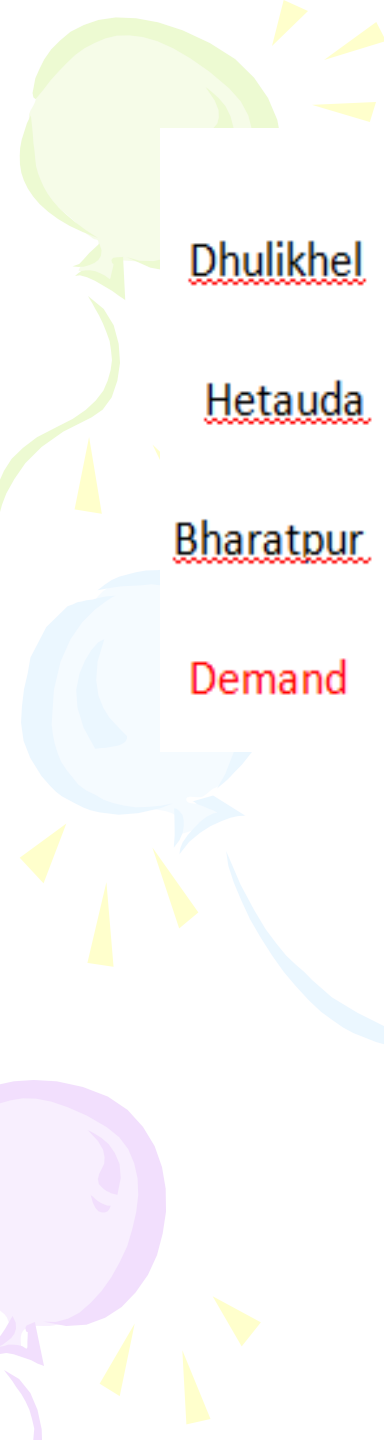


|                  | Civil Mall | Kathmandu Mall | People's Plaza | Supply  |
|------------------|------------|----------------|----------------|---------|
| <u>Dhulikhel</u> | 2          | 3              | 4              | 10      |
| <u>Hetauda</u>   | 11         | 10             | 2              | 20      |
| <u>Bharatpur</u> | 4          | 5              | 6              | 30      |
| Demand           | 15         | 30             | 15             | 60 / 60 |





|                  | Civil Mall | Kathmandu Mall | People's Plaza | Supply  |
|------------------|------------|----------------|----------------|---------|
| <u>Dhulikhel</u> | 2          | 3              | 4              | 10      |
| <u>Hetauda</u>   | 11         | 10             | 2              | 20      |
| <u>Bharatpur</u> | 4          | 5              | 6              | 30      |
| Demand           | 15         | 30             | 15             | 60 / 60 |



|                  | Civil Mall | Kathmandu Mall | People's Plaza | Supply  |
|------------------|------------|----------------|----------------|---------|
| <u>Dhulikhel</u> | 2          | 3              | 4              | 10      |
| <u>Hetauda</u>   | 11         | 10             | 2              | 20      |
| <u>Bharatpur</u> | 4          | 5              | 6              | 30      |
| Demand           | 15         | 30             | 15             | 60 / 60 |



**Example:** Solve the Transportation Table to find Initial Basic Feasible Solution using North-West Corner Method.

|                | D1 |  | D2 |  | D3 |  | D4 |  | Supply |
|----------------|----|--|----|--|----|--|----|--|--------|
| S <sub>1</sub> | 19 |  | 30 |  | 50 |  | 10 |  | 7      |
| S <sub>2</sub> | 70 |  | 30 |  | 40 |  | 60 |  | 9      |
| S <sub>3</sub> | 40 |  | 8  |  | 70 |  | 20 |  | 18     |
| Demand         | 5  |  | 8  |  | 7  |  | 14 |  | 34     |

**Example:** Solve the Transportation Table to find Initial Basic Feasible Solution using North-West Corner Method.

|        | D1 |  | D2 |  | D3 |  | D4 |  | Supply |
|--------|----|--|----|--|----|--|----|--|--------|
| S1     | 19 |  | 30 |  | 50 |  | 10 |  | 7      |
|        |    |  |    |  |    |  |    |  |        |
| S2     | 70 |  | 30 |  | 40 |  | 60 |  | 9      |
|        |    |  |    |  |    |  |    |  |        |
| S3     | 40 |  | 8  |  | 70 |  | 20 |  | 18     |
|        |    |  |    |  |    |  |    |  |        |
| Demand | 5  |  | 8  |  | 7  |  | 14 |  | 34     |

**Example:** Solve the Transportation Table to find Initial Basic Feasible Solution using North-West Corner Method.

|        | D1 |  | D2 |  | D3 |  | D4 |  | Supply |
|--------|----|--|----|--|----|--|----|--|--------|
| S1     | 19 |  | 30 |  | 50 |  | 10 |  | 7      |
|        |    |  |    |  |    |  |    |  |        |
| S2     | 70 |  | 30 |  | 40 |  | 60 |  | 9      |
|        |    |  |    |  |    |  |    |  |        |
| S3     | 40 |  | 8  |  | 70 |  | 20 |  | 18     |
|        |    |  |    |  |    |  |    |  |        |
| Demand | 5  |  | 8  |  | 7  |  | 14 |  | 34     |

**Example:** Solve the Transportation Table to find Initial Basic Feasible Solution using North-West Corner Method.

|        | D1 |  | D2 |  | D3 |  | D4 |  | Supply |
|--------|----|--|----|--|----|--|----|--|--------|
| S1     | 19 |  | 30 |  | 50 |  | 10 |  | 7      |
|        |    |  |    |  |    |  |    |  |        |
| S2     | 70 |  | 30 |  | 40 |  | 60 |  | 9      |
|        |    |  |    |  |    |  |    |  |        |
| S3     | 40 |  | 8  |  | 70 |  | 20 |  | 18     |
|        |    |  |    |  |    |  |    |  |        |
| Demand | 5  |  | 8  |  | 7  |  | 14 |  | 34     |

**Example:** Solve the Transportation Table to find Initial Basic Feasible Solution using North-West Corner Method.

$$\begin{aligned}\text{Total Cost} &= 19 \times 5 + 30 \times 2 + 30 \times 6 + 40 \times 3 + 70 \times 4 + 20 \times 14 \\ &= \text{Rs. } 1015\end{aligned}$$

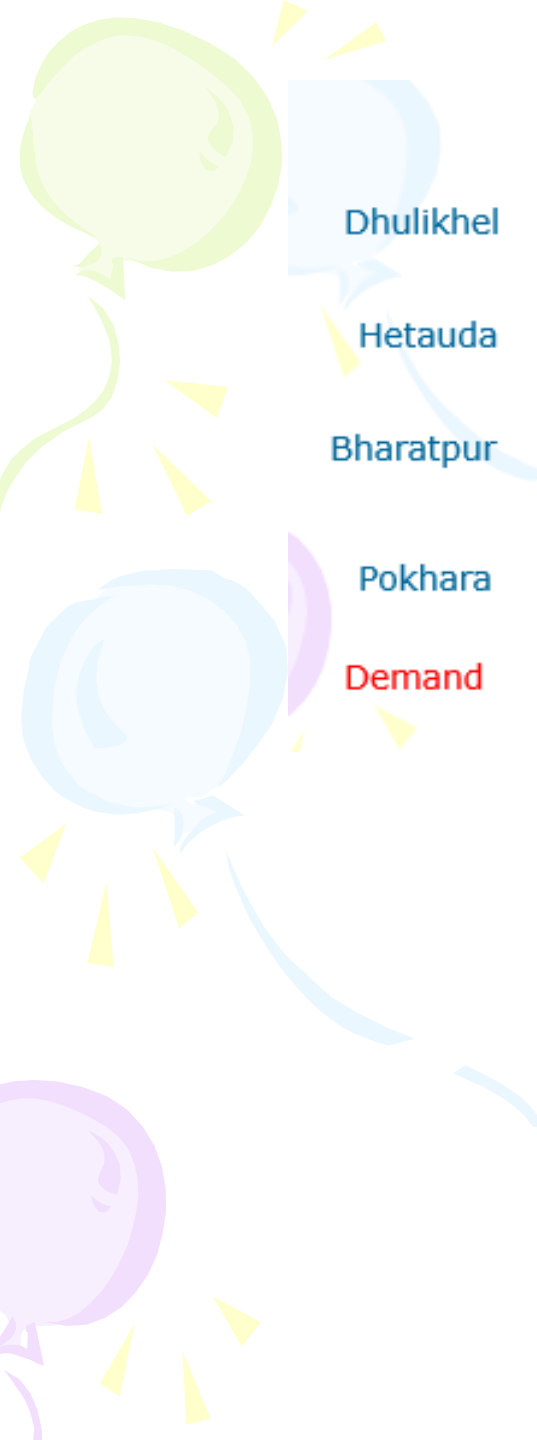
Feasibility: No. of allocations =  $m+n-1$   
 $6 = 3+4-1$   
 $6 = 6$

|        | D1 |   | D2 |   | D3 |   | D4 |    | Supply |
|--------|----|---|----|---|----|---|----|----|--------|
| S1     | 19 |   | 30 |   | 50 |   | 10 |    | 7      |
|        |    | 5 |    | 2 |    |   |    |    |        |
| S2     | 70 |   | 30 |   | 40 |   | 60 |    | 9      |
|        |    |   |    | 6 |    | 3 |    |    |        |
| S3     | 40 |   | 8  |   | 70 |   | 20 |    | 18     |
|        |    |   |    |   |    | 4 |    | 14 |        |
| Demand | 5  |   | 8  |   | 7  |   | 14 |    | 34     |

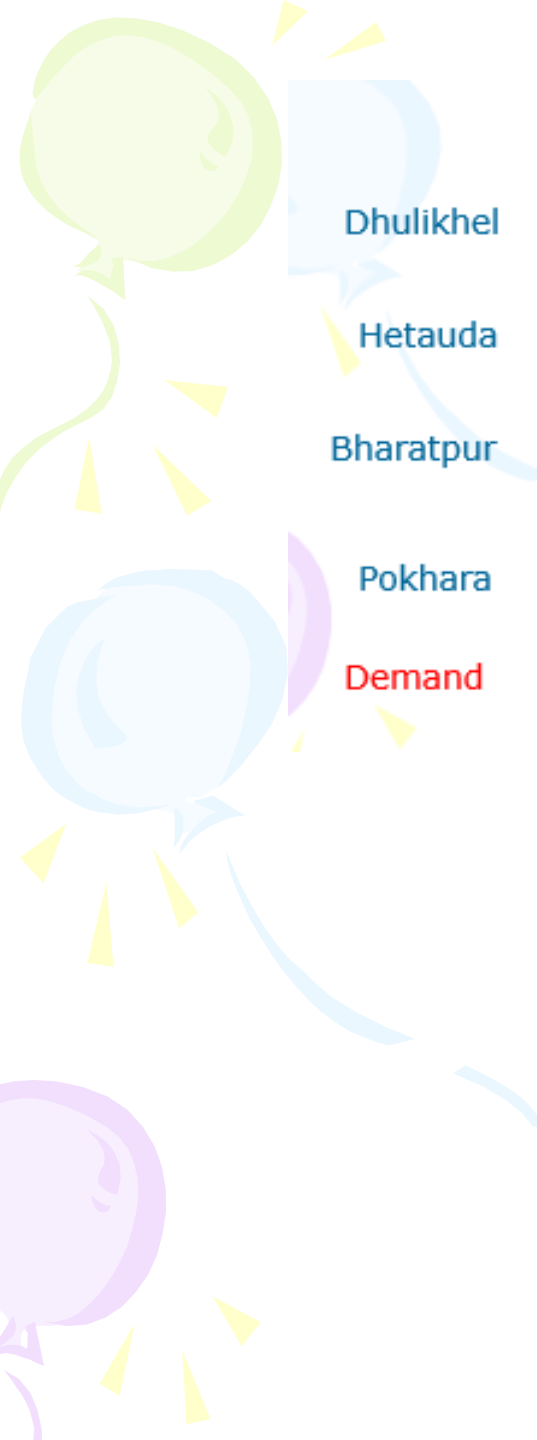


# Problem

|           | Civil Mall | Kathmandu Mall | People's Plaza | Supply |
|-----------|------------|----------------|----------------|--------|
| Dhulikhel | 3          | 2              | 1              | 20     |
| Hetauda   | 2          | 4              | 1              | 40     |
| Bharatpur | 3          | 5              | 2              | 40     |
| Pokhara   | 4          | 6              | 7              | 25     |
| Demand    | 40         | 30             | 55             |        |



|           | Civil Mall | Kathmandu Mall | People's Plaza | Supply |
|-----------|------------|----------------|----------------|--------|
| Dhulikhel | 3          | 2              | 1              | 20     |
| Hetauda   | 2          | 4              | 1              | 40     |
| Bharatpur | 3          | 5              | 2              | 40     |
| Pokhara   | 4          | 6              | 7              | 25     |
| Demand    | 40         | 30             | 55             |        |



|           | Civil Mall | Kathmandu Mall | People's Plaza | Supply |
|-----------|------------|----------------|----------------|--------|
| Dhulikhel | 3          | 2              | 1              | 20     |
| Hetauda   | 2          | 4              | 1              | 40     |
| Bharatpur | 3          | 5              | 2              | 40     |
| Pokhara   | 4          | 6              | 7              | 25     |
| Demand    | 40         | 30             | 55             |        |

# Problem

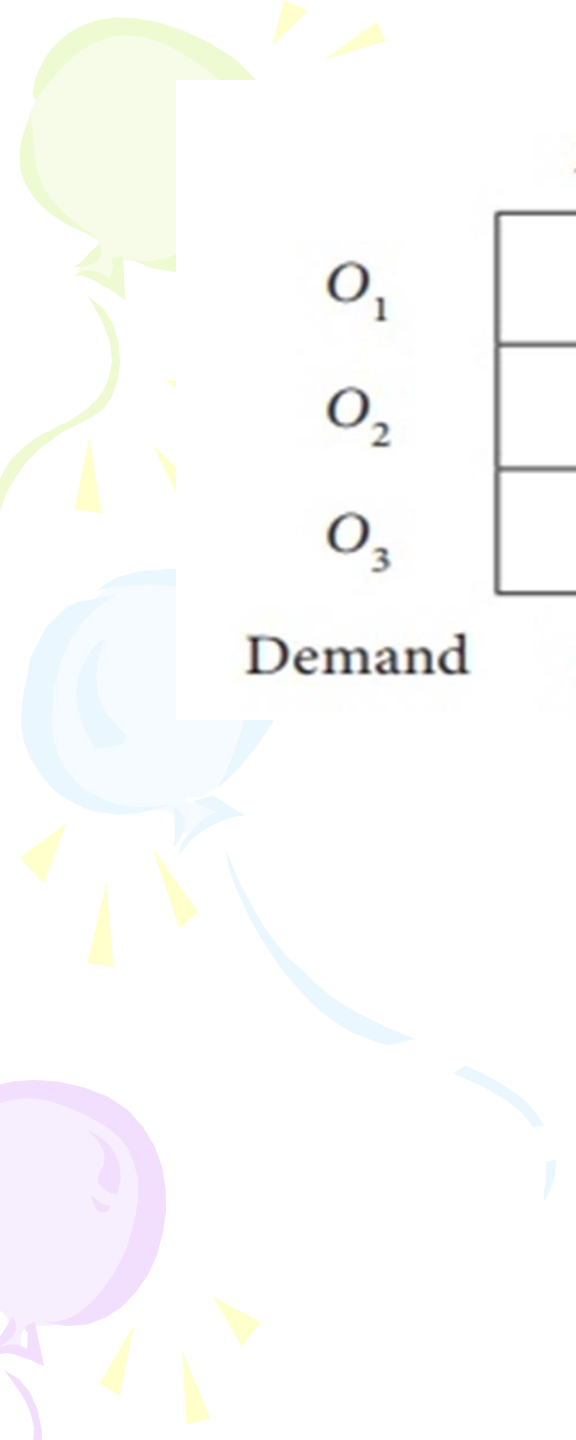
Total Cost :  $20 \times 3 + 20 \times 2 + 20 \times 4 + 10 \times 5 + 30 \times 2 + 25 \times 7 = 465$

BFS: No. of allocations =  $m+n-1$   
 $6 = 4+3-1=6$

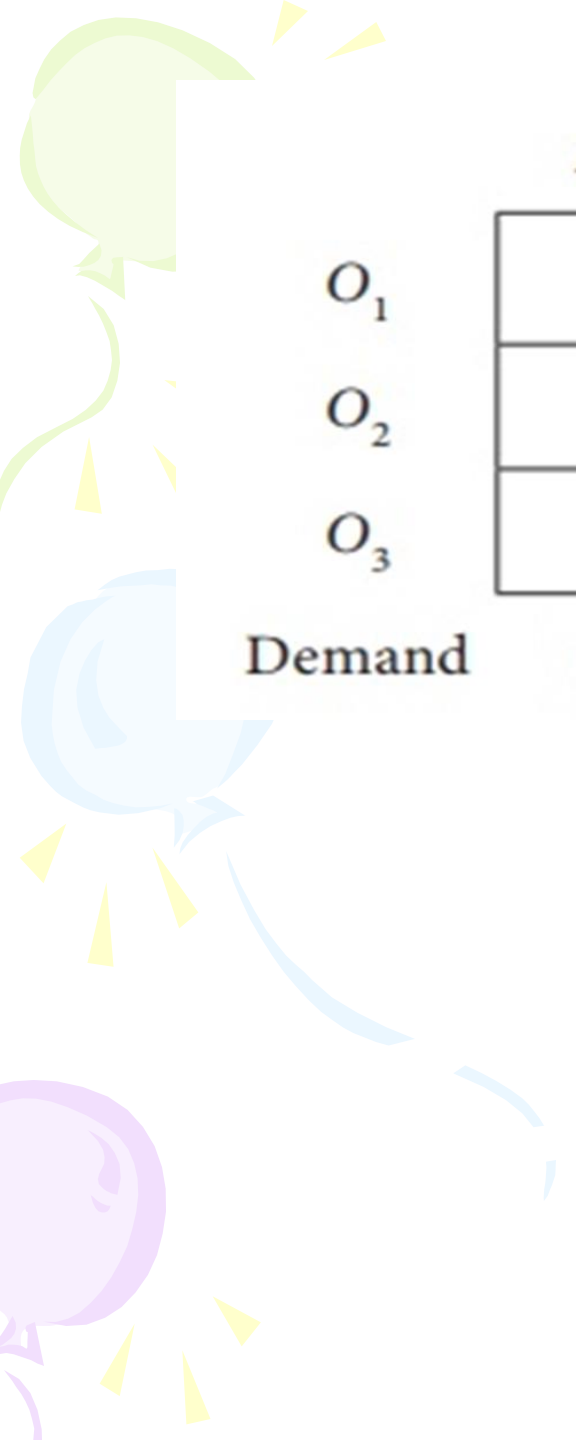
|           | Civil Mall | Kathmandu Mall | People's Plaza | Supply |
|-----------|------------|----------------|----------------|--------|
| Dhulikhel | 20 3       | 2              | 1              | 20     |
| Hetauda   | 20 2       | 20 4           | 1              | 40     |
| Bharatpur | 3          | 10 5           | 30 2           | 40     |
| Pokhara   | 4          | 6              | 25 7           | 25     |
| Demand    | 40         | 30             | 55             | 125    |

# Problem

|        | $D_1$ | $D_2$ | $D_3$ | $D_4$ | Supply |
|--------|-------|-------|-------|-------|--------|
| $O_1$  | 5     | 3     | 6     | 2     | 19     |
| $O_2$  | 4     | 7     | 9     | 1     | 37     |
| $O_3$  | 3     | 4     | 7     | 5     | 34     |
| Demand | 16    | 18    | 31    | 25    |        |



|        | $D_1$ | $D_2$ | $D_3$ | $D_4$ | Supply |
|--------|-------|-------|-------|-------|--------|
| $O_1$  | 5     | 3     | 6     | 2     | 19     |
| $O_2$  | 4     | 7     | 9     | 1     | 37     |
| $O_3$  | 3     | 4     | 7     | 5     | 34     |
| Demand | 16    | 18    | 31    | 25    |        |



|        | $D_1$ | $D_2$ | $D_3$ | $D_4$ | Supply |
|--------|-------|-------|-------|-------|--------|
| $O_1$  | 5     | 3     | 6     | 2     | 19     |
| $O_2$  | 4     | 7     | 9     | 1     | 37     |
| $O_3$  | 3     | 4     | 7     | 5     | 34     |
| Demand | 16    | 18    | 31    | 25    |        |

# Least Cost Method

**Step1:** Select the cell having lowest unit cost in the entire table and allocate the minimum of supply or demand values in that cell.

**Step2:** Then eliminate the row or column in which supply or demand is exhausted. If both the supply and demand values are same, either of the row or column can be eliminated.

In case, the smallest unit cost is not unique, then select the cell where maximum allocation can be made.

**Step3:** Repeat the process with next lowest unit cost and continue until the entire available supply at various sources and demand at various destinations is satisfied.



**Example:** Solve the Transportation Table to find Initial Basic Feasible Solution using North-West Corner Method.

|        | D1 |  | D2 |  | D3 |  | D4 |  | Supply |
|--------|----|--|----|--|----|--|----|--|--------|
| S1     | 19 |  | 30 |  | 50 |  | 10 |  | 7      |
|        |    |  |    |  |    |  |    |  |        |
| S2     | 70 |  | 30 |  | 40 |  | 60 |  | 9      |
|        |    |  |    |  |    |  |    |  |        |
| S3     | 40 |  | 8  |  | 70 |  | 20 |  | 18     |
|        |    |  |    |  |    |  |    |  |        |
| Demand | 5  |  | 8  |  | 7  |  | 14 |  | 34     |

**Example:** Solve the Transportation Table to find Initial Basic Feasible Solution using North-West Corner Method.

|        | D1 |  | D2 |  | D3 |  | D4 |  | Supply |
|--------|----|--|----|--|----|--|----|--|--------|
| S1     | 19 |  | 30 |  | 50 |  | 10 |  | 7      |
|        |    |  |    |  |    |  |    |  |        |
| S2     | 70 |  | 30 |  | 40 |  | 60 |  | 9      |
|        |    |  |    |  |    |  |    |  |        |
| S3     | 40 |  | 8  |  | 70 |  | 20 |  | 18     |
|        |    |  |    |  |    |  |    |  |        |
| Demand | 5  |  | 8  |  | 7  |  | 14 |  | 34     |

**Example:** Solve the Transportation Table to find Initial Basic Feasible Solution using North-West Corner Method.

|        | D1 |  | D2 |  | D3 |  | D4 |  | Supply |
|--------|----|--|----|--|----|--|----|--|--------|
| S1     | 19 |  | 30 |  | 50 |  | 10 |  | 7      |
|        |    |  |    |  |    |  |    |  |        |
| S2     | 70 |  | 30 |  | 40 |  | 60 |  | 9      |
|        |    |  |    |  |    |  |    |  |        |
| S3     | 40 |  | 8  |  | 70 |  | 20 |  | 18     |
|        |    |  |    |  |    |  |    |  |        |
| Demand | 5  |  | 8  |  | 7  |  | 14 |  | 34     |

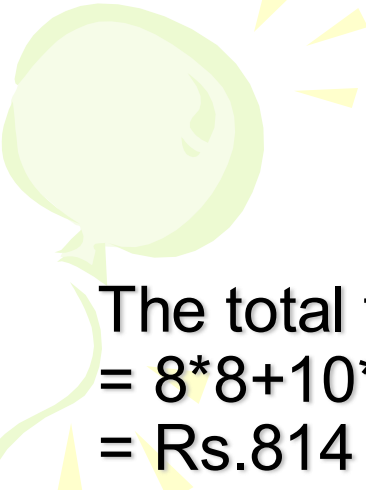
|        | D1 | D2 | D3 | D4 | Supply |
|--------|----|----|----|----|--------|
| S1     | 19 | 30 | 50 | 10 | 7      |
| S2     | 70 | 30 | 40 | 60 | 9      |
| S3     | 40 | 8  | 70 | 20 | 18     |
| Demand | 5  | 8  | 7  | 14 | 34     |

|        | D1 | D3 | D4 | Supply |
|--------|----|----|----|--------|
| S1     | 19 | 50 | 10 | 7      |
| S2     | 70 | 40 | 60 | 9      |
| S3     | 40 | 70 | 20 | 10     |
| Demand | 5  | 7  | 14 | 34     |

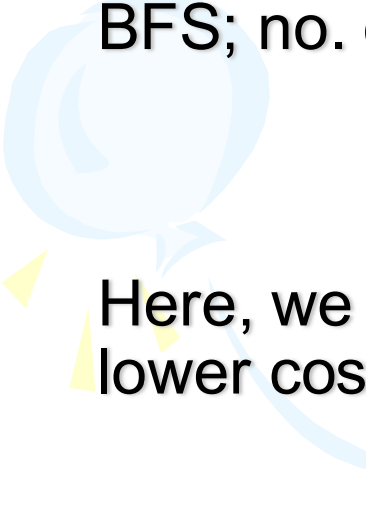
|        | D1 | D3 | D4 | Supply |
|--------|----|----|----|--------|
| S2     | 70 | 40 | 60 | 9      |
| S3     | 40 | 70 | 20 | 10     |
| Demand | 5  | 7  | 7  | 34     |

|        | D1 | D3 | Supply |
|--------|----|----|--------|
| S2     | 70 | 40 | 9      |
| S3     | 40 | 70 | 3      |
| Demand | 5  | 7  | 34     |

|        | D1 | Supply |
|--------|----|--------|
| S2     | 70 | 2      |
| S3     | 40 | 3      |
| Demand | 5  | 34     |



The total transportation cost obtained by this method  
=  $8 \times 8 + 10 \times 7 + 20 \times 7 + 40 \times 7 + 70 \times 2 + 40 \times 3$   
= Rs.814



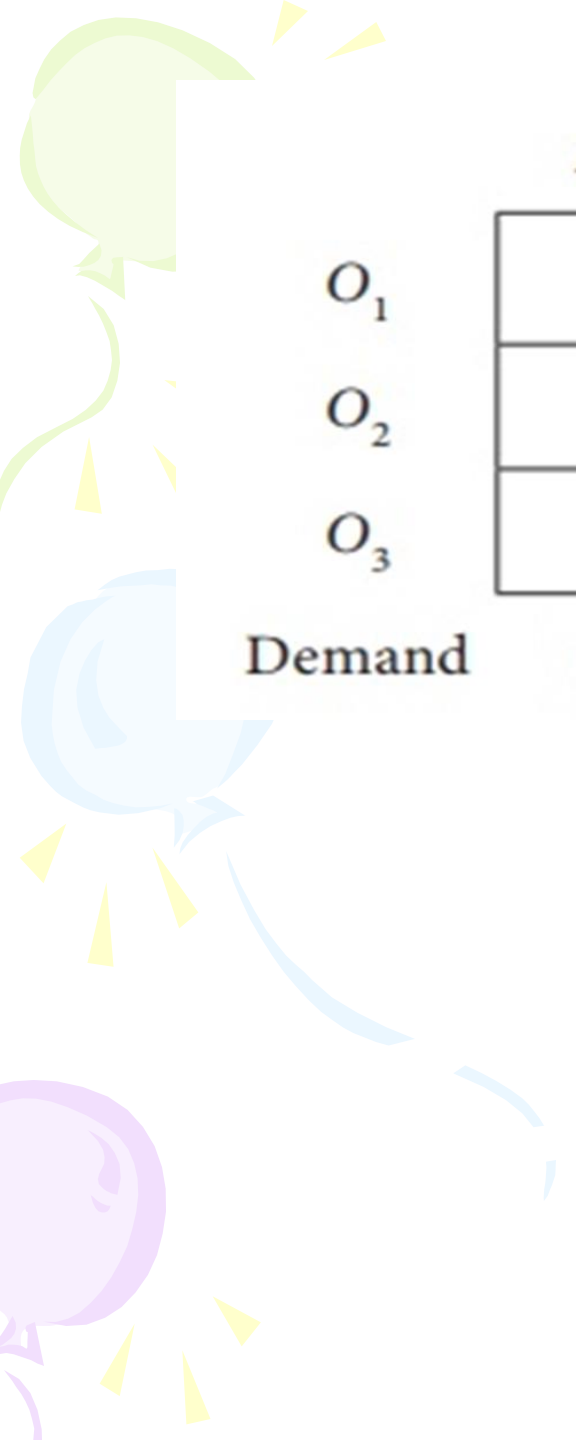
BFS; no. of allocations =  $m+n-1$   
 $6 = 3+4-1$   
 $6 = 6$



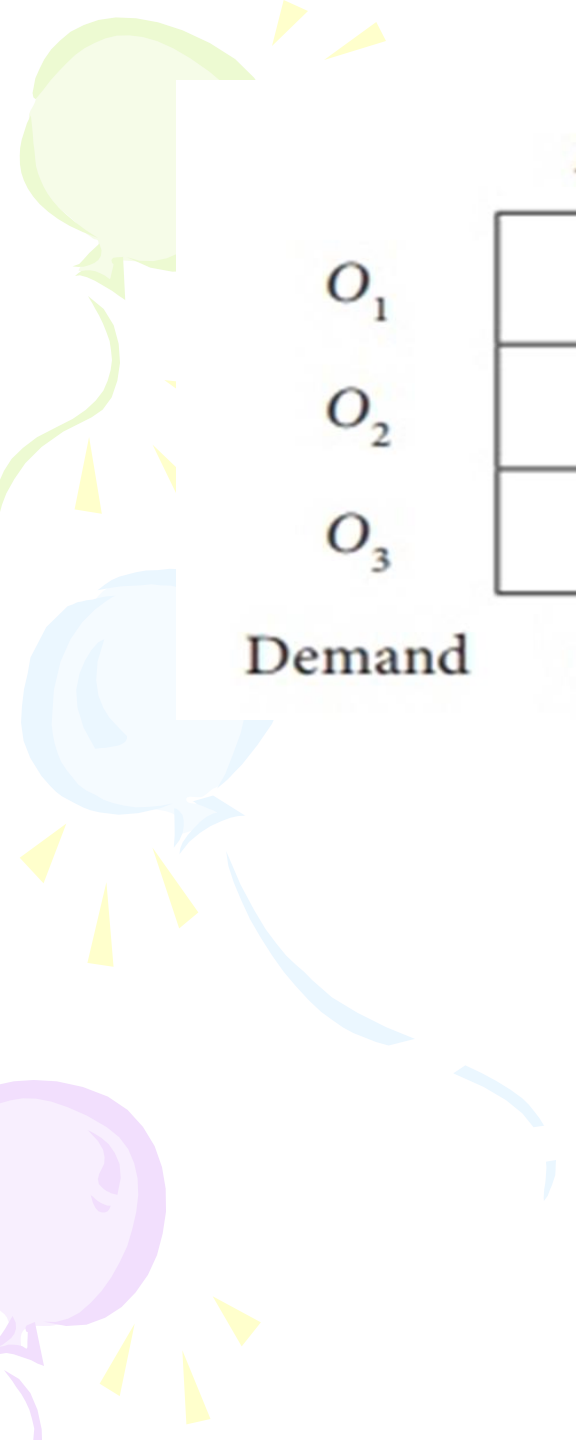
Here, we can see that the ***Least Cost Method*** involves a lower cost than the *North-West Corner Method*.

# Problem 2

|        | $D_1$ | $D_2$ | $D_3$ | $D_4$ | Supply |
|--------|-------|-------|-------|-------|--------|
| $O_1$  | 5     | 3     | 6     | 2     | 19     |
| $O_2$  | 4     | 7     | 9     | 1     | 37     |
| $O_3$  | 3     | 4     | 7     | 5     | 34     |
| Demand | 16    | 18    | 31    | 25    |        |



|        | $D_1$ | $D_2$ | $D_3$ | $D_4$ | Supply |
|--------|-------|-------|-------|-------|--------|
| $O_1$  | 5     | 3     | 6     | 2     | 19     |
| $O_2$  | 4     | 7     | 9     | 1     | 37     |
| $O_3$  | 3     | 4     | 7     | 5     | 34     |
| Demand | 16    | 18    | 31    | 25    |        |



|        | $D_1$ | $D_2$ | $D_3$ | $D_4$ | Supply |
|--------|-------|-------|-------|-------|--------|
| $O_1$  | 5     | 3     | 6     | 2     | 19     |
| $O_2$  | 4     | 7     | 9     | 1     | 37     |
| $O_3$  | 3     | 4     | 7     | 5     | 34     |
| Demand | 16    | 18    | 31    | 25    |        |



# Problem 2

Total cost =  $18 \times 3 + 1 \times 6 + 12 \times 9 + 25 \times 1 + 16 \times 3 + 18 \times 7 = 367$

BFS: No. Allocations =  $m+n-1$

$$6 = 3+4-1=6$$

|        | $D_1$              | $D_2$              | $D_3$               | $D_4$              | Supply            |
|--------|--------------------|--------------------|---------------------|--------------------|-------------------|
| $O_1$  | 5                  | 18<br>3            | 1<br>6              | 2                  | <del>19</del> 1 0 |
| $O_2$  | 4                  | 7                  | 12<br>9             | 25<br>1            | <del>37</del> 12  |
| $O_3$  | 16<br>3            | 4                  | 18<br>7             | 5                  | <del>34</del> 18  |
| Demand | <del>16</del><br>0 | <del>18</del><br>0 | <del>31</del><br>30 | <del>25</del><br>0 |                   |

# Vogel's Approximation Method

**Step1:** Calculate penalty for each row and column by taking the difference between the two smallest unit costs. This penalty or extra cost has to be paid if one fails to allocate the minimum unit transportation cost.

**Step2:** Select the row or column with the highest penalty and select the minimum unit cost of that row or column. Then, allocate the minimum of supply or demand values in that cell. If there is a tie, then select the cell where maximum allocation could be made.

**Step3:** Adjust the supply and demand and eliminate the satisfied row or column. If a row and column are satisfied simultaneously, only one of them is eliminated and the other one is assigned a **zero** value. Any row or column having **zero** supply or demand, can not be used in calculating future penalties.

**Step4:** Repeat the process until all the supply sources and demand destinations are satisfied.

**Example:** Solve the Transportation Table to find Initial Basic Feasible Solution using North-West Corner Method.

|        | D1 |  | D2 |  | D3 |  | D4 |  | Supply |
|--------|----|--|----|--|----|--|----|--|--------|
| S1     | 19 |  | 30 |  | 50 |  | 10 |  | 7      |
|        |    |  |    |  |    |  |    |  |        |
| S2     | 70 |  | 30 |  | 40 |  | 60 |  | 9      |
|        |    |  |    |  |    |  |    |  |        |
| S3     | 40 |  | 8  |  | 70 |  | 20 |  | 18     |
|        |    |  |    |  |    |  |    |  |        |
| Demand | 5  |  | 8  |  | 7  |  | 14 |  | 34     |

**Example:** Solve the Transportation Table to find Initial Basic Feasible Solution using North-West Corner Method.

|        | D1 |  | D2 |  | D3 |  | D4 |  | Supply |
|--------|----|--|----|--|----|--|----|--|--------|
| S1     | 19 |  | 30 |  | 50 |  | 10 |  | 7      |
|        |    |  |    |  |    |  |    |  |        |
| S2     | 70 |  | 30 |  | 40 |  | 60 |  | 9      |
|        |    |  |    |  |    |  |    |  |        |
| S3     | 40 |  | 8  |  | 70 |  | 20 |  | 18     |
|        |    |  |    |  |    |  |    |  |        |
| Demand | 5  |  | 8  |  | 7  |  | 14 |  | 34     |

**Example:** Solve the Transportation Table to find Initial Basic Feasible Solution using North-West Corner Method.

|        | D1 |  | D2 |  | D3 |  | D4 |  | Supply |
|--------|----|--|----|--|----|--|----|--|--------|
| S1     | 19 |  | 30 |  | 50 |  | 10 |  | 7      |
|        |    |  |    |  |    |  |    |  |        |
| S2     | 70 |  | 30 |  | 40 |  | 60 |  | 9      |
|        |    |  |    |  |    |  |    |  |        |
| S3     | 40 |  | 8  |  | 70 |  | 20 |  | 18     |
|        |    |  |    |  |    |  |    |  |        |
| Demand | 5  |  | 8  |  | 7  |  | 14 |  | 34     |

|        | D1 | D2 | D3 | D4 | Supply |
|--------|----|----|----|----|--------|
| S1     | 19 | 30 | 50 | 10 | 7      |
| S2     | 70 | 30 | 40 | 60 | 9      |
| S3     | 40 | 8  | 70 | 20 | 18     |
| Demand | 5  | 8  | 7  | 14 | 34     |



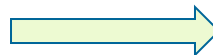
|         | D1 | D2 | D3 | D4 | Supply | Penalty |
|---------|----|----|----|----|--------|---------|
| S1      | 19 | 30 | 50 | 10 | 7      | 9       |
| S2      | 70 | 30 | 40 | 60 | 9      | 10      |
| S3      | 40 | 8  | 70 | 20 | 10     | 12      |
| Demand  | 5  | 0  | 7  | 14 |        |         |
| penalty | 21 | 22 | 10 | 10 |        |         |

|        | D1 | D3 | D4 | Supply |
|--------|----|----|----|--------|
| S1     | 19 | 50 | 10 | 7      |
| S2     | 70 | 40 | 60 | 9      |
| S3     | 40 | 70 | 20 | 10     |
| Demand | 5  | 7  | 14 |        |



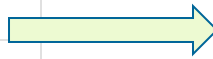
|         | D1 | D3 | D4 | Supply | Penalty |
|---------|----|----|----|--------|---------|
| S1      | 19 | 50 | 10 | 2      | 9       |
| S2      | 70 | 40 | 60 | 9      | 20      |
| S3      | 40 | 70 | 20 | 10     | 20      |
| Demand  | 0  | 7  | 14 |        |         |
| Penalty | 21 | 10 | 10 |        |         |

|        | D3 | D4 | Supply |
|--------|----|----|--------|
| S1     | 50 | 10 | 2      |
| S2     | 40 | 60 | 9      |
| S3     | 70 | 20 | 10     |
| Demand | 7  | 14 |        |



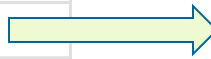
|         | D3 | D4 | Supply | Penalty |
|---------|----|----|--------|---------|
| S1      | 50 | 10 | 2      | 40      |
| S2      | 40 | 60 | 9      | 20      |
| S3      | 70 | 20 | 0      | 50      |
| Demand  | 7  | 4  |        |         |
| Penalty | 10 | 10 |        |         |

|        | D3 | D4 | Supply |
|--------|----|----|--------|
| S1     | 50 | 10 | 2      |
| S2     | 40 | 60 | 9      |
| Demand | 7  | 4  |        |



|         | D3 | D4 | Supply | Penalty |
|---------|----|----|--------|---------|
| S1      | 50 | 10 | 0      | 40      |
| S2      | 40 | 60 | 9      | 20      |
| Demand  | 7  | 2  |        |         |
| Penalty | 10 | 50 |        |         |

|         | D3 | D4 | Supply |
|---------|----|----|--------|
| S2      | 40 | 60 | 9      |
| Demand  | 7  | 2  |        |
| Penalty |    |    |        |



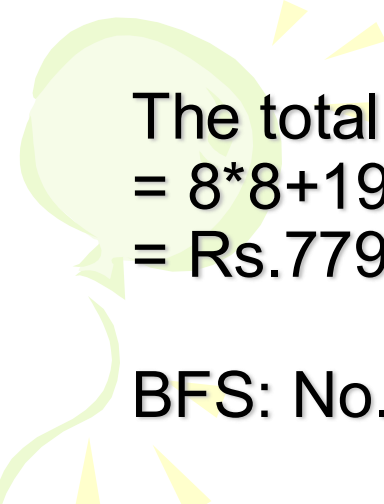
|         | D3 | D4 | Supply |
|---------|----|----|--------|
| S2      | 40 | 60 | 2      |
| Demand  | 0  | 2  |        |
| Penalty |    |    |        |

|         | D4 | Supply |
|---------|----|--------|
| S2      | 60 | 0      |
| Demand  | 0  |        |
| Penalty |    |        |

The total transportation cost =  $8 \times 8 + 19 \times 5 + 20 \times 10 + 10 \times 2 + 40 \times 7 + 60 \times 2 = \text{Rs.}779$

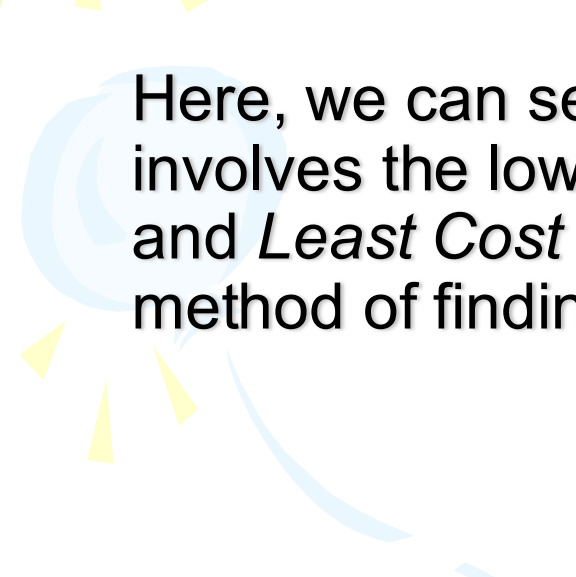
BFS: No. of allocations =  $m+n-1$

$$6 = 3 + 4 - 1 = 6$$



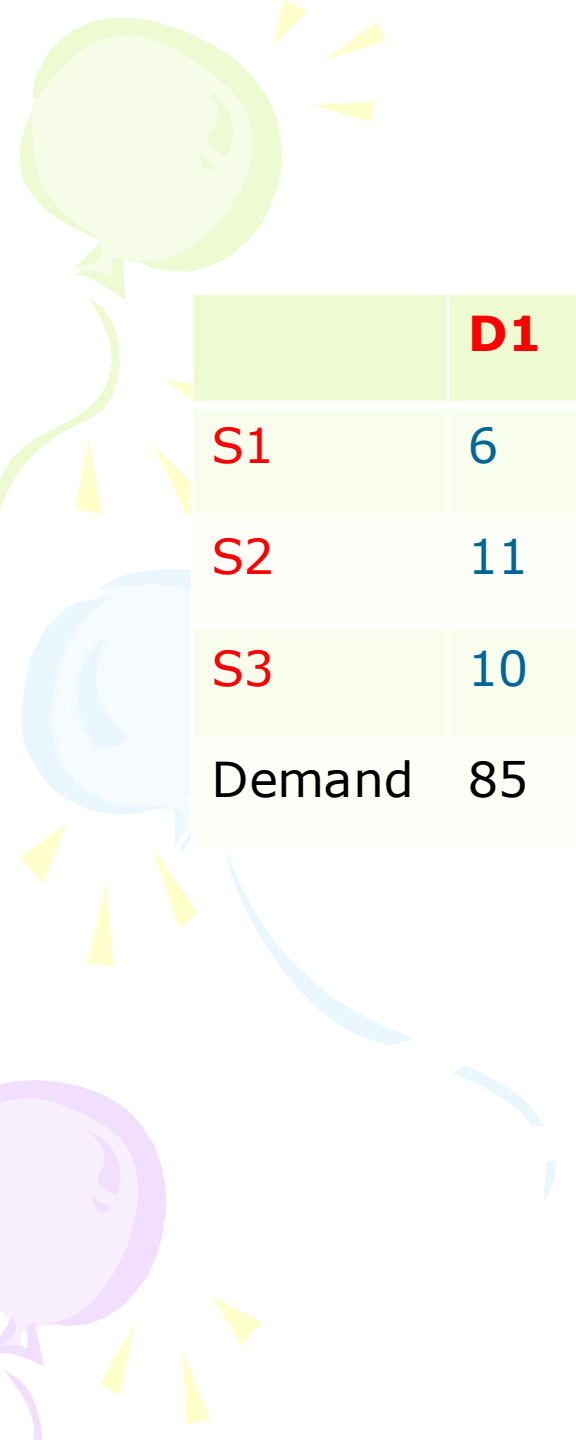
The total transportation cost obtained by this method  
=  $8*8+19*5+20*10+10*2+40*7+60*2$   
= Rs.779

BFS: No. of allocations =  $m+n-1$   
 $6=3+4-1=6$

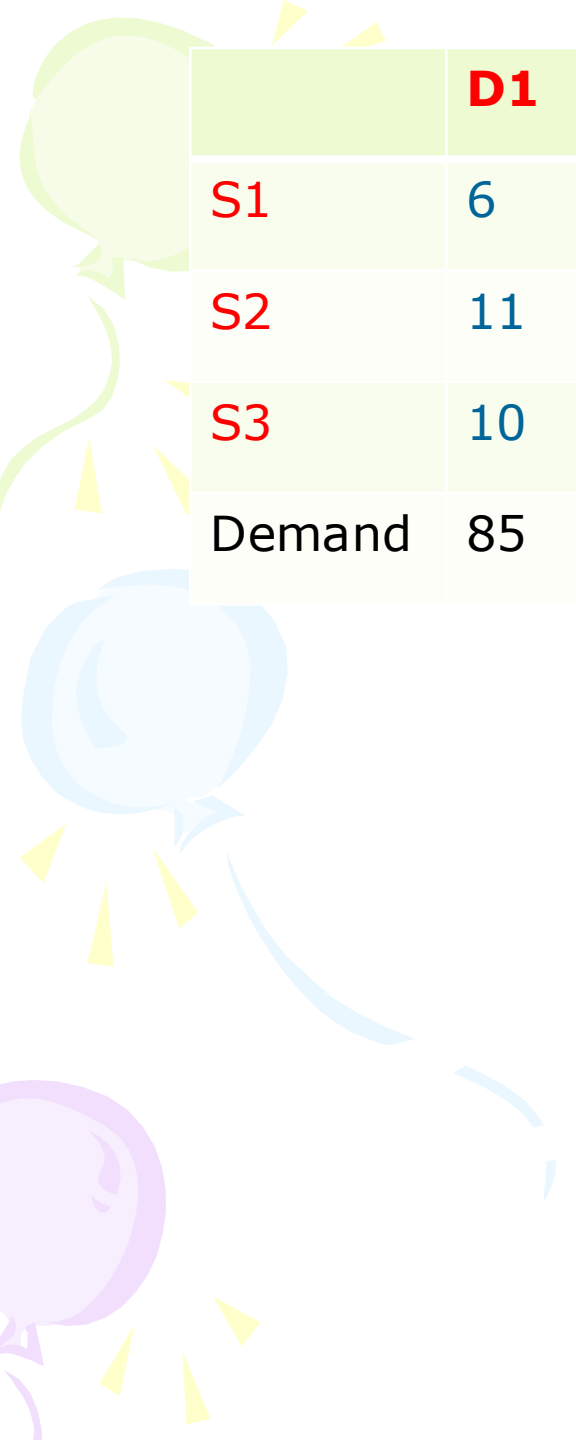


Here, we can see that ***Vogel's Approximation Method*** involves the lowest cost than *North-West Corner Method* and *Least Cost Method* and hence is the most preferred method of finding initial basic feasible solution.

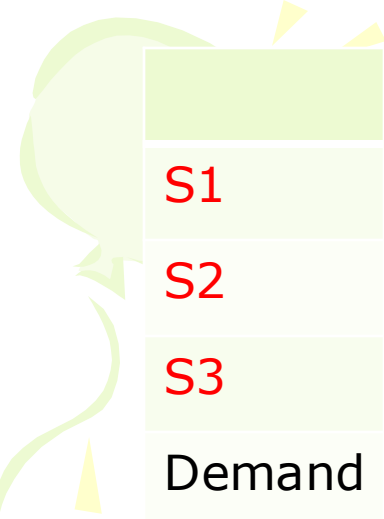




|           | <b>D1</b> | <b>D2</b> | <b>D3</b> | <b>D4</b> | <b>Supply</b> |
|-----------|-----------|-----------|-----------|-----------|---------------|
| <b>S1</b> | 6         | 1         | 9         | 3         | 70            |
| <b>S2</b> | 11        | 5         | 2         | 8         | 55            |
| <b>S3</b> | 10        | 12        | 4         | 7         | 70            |
| Demand    | 85        | 35        | 50        | 45        |               |



|        | D1 | D2 | D3 | D4 | Supply |
|--------|----|----|----|----|--------|
| S1     | 6  | 1  | 9  | 3  | 70     |
| S2     | 11 | 5  | 2  | 8  | 55     |
| S3     | 10 | 12 | 4  | 7  | 70     |
| Demand | 85 | 35 | 50 | 45 |        |



|        | D1 | D2 | D3 | D4 | Supply |
|--------|----|----|----|----|--------|
| S1     | 6  | 1  | 9  | 3  | 70     |
| S2     | 11 | 5  | 2  | 8  | 55     |
| S3     | 10 | 12 | 4  | 7  | 70     |
| Demand | 85 | 35 | 50 | 45 |        |





# Unbalanced Transportation problems

Transportation  
problems

Balance Problem

$$\sum Supplies = \sum Demands$$

Unbalanced problem

$$\sum Supplies \neq \sum Demands$$

$$\sum Supplies > \sum Demands$$

1. Add one additional dummy demand point (or Column)
2. Dummy column value shall be  $(\sum Supplies - \sum Demands)$  and, for its each cell, assume 'zero' unit transportation cost

$$\sum Supplies < \sum Demands$$

1. Add one additional dummy supply point (or Row)
2. Dummy row value shall be  $(\sum Demands - \sum Supplies)$  and, for its each cell, assume 'zero' unit transportation cost



# Unbalanced Transportation problems

Example of unbalanced problem

|           | <b>D1</b> | <b>D2</b> | <b>D3</b> | <b>D4</b> | Supply    |
|-----------|-----------|-----------|-----------|-----------|-----------|
| <b>S1</b> | 6         | 1         | 9         | 3         | <b>70</b> |
| <b>S2</b> | 11        | 5         | 2         | 8         | <b>55</b> |
| <b>S3</b> | 10        | 12        | 4         | 7         | <b>70</b> |
| Demand    | <b>85</b> | <b>35</b> | <b>50</b> | <b>45</b> |           |

$$\begin{aligned}\sum \text{Supplies} &= 70 + 55 + 70 = 195 \\ \sum \text{Demands} &= 85 + 35 + 50 + 45 = 215 \\ \sum \text{Demands} &> \sum \text{Supplies}\end{aligned}$$

$$\text{Excess} = \sum \text{Demands} - \sum \text{Supplies} = 215 - 195 = 20$$

Add 'dummy supply point' (or extra row) with supply capacity 20.

# Unbalanced Transportation problems

|        | D1 | D2 | D3 | D4 | Supply |
|--------|----|----|----|----|--------|
| S1     | 6  | 1  | 9  | 3  | 70     |
| S2     | 11 | 5  | 2  | 8  | 55     |
| S3     | 10 | 12 | 4  | 7  | 70     |
| Demand | 85 | 35 | 50 | 45 |        |

|        | D1 | D2 | D3 | D4 | Supply |
|--------|----|----|----|----|--------|
| S1     | 6  | 1  | 9  | 3  | 70     |
| S2     | 11 | 5  | 2  | 8  | 55     |
| S3     | 10 | 12 | 4  | 7  | 70     |
| S4     | 0  | 0  | 0  | 0  | 20     |
| Demand | 85 | 35 | 50 | 45 |        |

Now,  $\sum \text{Demands} = \sum \text{Supplies}, 215 = 215$

# Unbalanced Transportation problems

## North-West corner method

|           | <b>D1</b> | <b>D2</b> | <b>D3</b> | <b>D4</b> | Supply    |
|-----------|-----------|-----------|-----------|-----------|-----------|
| <b>S1</b> | 6         | 1         | 9         | 3         | <b>70</b> |
| <b>S2</b> | 11        | 5         | 2         | 8         | <b>55</b> |
| <b>S3</b> | 10        | 12        | 4         | 7         | <b>70</b> |
| <b>S4</b> | 0         | 0         | 0         | 0         | <b>20</b> |
| Demand    | <b>85</b> | <b>35</b> | <b>50</b> | <b>45</b> |           |

# Unbalanced Transportation problems

|           | <b>D1</b> | <b>D2</b> | <b>D3</b> | <b>D4</b> | Supply    |
|-----------|-----------|-----------|-----------|-----------|-----------|
| <b>S1</b> | 6         | 1         | 9         | 3         | <b>70</b> |
| <b>S2</b> | 11        | 5         | 2         | 8         | <b>55</b> |
| <b>S3</b> | 10        | 12        | 4         | 7         | <b>70</b> |
| <b>S4</b> | 0         | 0         | 0         | 0         | <b>20</b> |
| Demand    | <b>85</b> | <b>35</b> | <b>50</b> | <b>45</b> |           |

|           | <b>D1</b> | <b>D2</b> | <b>D3</b> | <b>D4</b> | Supply    |
|-----------|-----------|-----------|-----------|-----------|-----------|
| <b>S2</b> | 11        | 5         | 2         | 8         | <b>55</b> |
| <b>S3</b> | 10        | 12        | 4         | 7         | <b>70</b> |
| <b>S4</b> | 0         | 0         | 0         | 0         | <b>20</b> |
| Demand    | <b>15</b> | <b>35</b> | <b>50</b> | <b>45</b> |           |

|           | <b>D2</b> | <b>D3</b> | <b>D4</b> | Supply    |
|-----------|-----------|-----------|-----------|-----------|
| <b>S2</b> | 5         | 2         | 8         | <b>40</b> |
| <b>S3</b> | 12        | 4         | 7         | <b>70</b> |
| <b>S4</b> | 0         | 0         | 0         | <b>20</b> |
| Demand    | <b>35</b> | <b>50</b> | <b>45</b> |           |



# Unbalanced Transportation problems

|        | D3 | D4 | Supply |
|--------|----|----|--------|
| S2     | 2  | 8  | 5      |
| S3     | 4  | 7  | 70     |
| S4     | 0  | 0  | 20     |
| Demand | 50 | 45 |        |

|        | D3 | D4 | Supply |
|--------|----|----|--------|
| S3     | 4  | 7  | 70     |
| S4     | 0  | 0  | 20     |
| Demand | 45 | 45 |        |

|        | D4 | Supply |
|--------|----|--------|
| S3     | 7  | 25     |
| S4     | 0  | 20     |
| Demand | 45 |        |



# Unbalanced Transportation problems

Least cost Method

|           | <b>D1</b> | <b>D2</b> | <b>D3</b> | <b>D4</b> | Supply    |
|-----------|-----------|-----------|-----------|-----------|-----------|
| <b>S1</b> | 6         | 1         | 9         | 3         | <b>70</b> |
| <b>S2</b> | 11        | 5         | 2         | 8         | <b>55</b> |
| <b>S3</b> | 10        | 12        | 4         | 7         | <b>70</b> |
| <b>S4</b> | 0         | 0         | 0         | 0         | <b>20</b> |
| Demand    | <b>85</b> | <b>35</b> | <b>50</b> | <b>45</b> |           |



# Unbalanced Transportation problems

|           | <b>D1</b> | <b>D2</b> | <b>D3</b> | <b>D4</b> | Supply    |
|-----------|-----------|-----------|-----------|-----------|-----------|
| <b>S1</b> | 6         | 1         | 9         | 3         | <b>70</b> |
| <b>S2</b> | 11        | 5         | 2         | 8         | <b>55</b> |
| <b>S3</b> | 10        | 12        | 4         | 7         | <b>70</b> |
| <b>S4</b> | 0         | 0         | 0         | 0         | <b>20</b> |
| Demand    | <b>85</b> | <b>35</b> | <b>50</b> | <b>45</b> |           |

|        | D1 | D2 | D3 | D4 | Supply |
|--------|----|----|----|----|--------|
| S1     | 6  | 1  | 9  | 3  | 70     |
| S2     | 11 | 5  | 2  | 8  | 55     |
| S3     | 10 | 12 | 4  | 7  | 70     |
| Demand | 85 | 35 | 50 | 25 |        |

|        | D1 | D3 | D4 | Supply |
|--------|----|----|----|--------|
| S1     | 6  | 9  | 3  | 35     |
| S2     | 11 | 2  | 8  | 55     |
| S3     | 10 | 4  | 7  | 70     |
| Demand | 85 | 50 | 25 |        |

# Unbalanced Transportation problems

|        |    |    |        |  |        |    |        |
|--------|----|----|--------|--|--------|----|--------|
|        | D1 | D4 | Supply |  |        | D1 | Supply |
| S1     | 6  | 3  | 35     |  | S1     | 6  | 10     |
| S2     | 11 | 8  | 5      |  | S2     | 11 | 5      |
| S3     | 10 | 7  | 70     |  | S3     | 10 | 70     |
| Demand | 85 | 25 |        |  | Demand | 85 |        |

|           | <b>D1</b> | Supply    |
|-----------|-----------|-----------|
| <b>S2</b> | 11        | <b>5</b>  |
| <b>S3</b> | 10        | <b>70</b> |
| Demand    | <b>75</b> |           |



# Unbalanced Transportation problems

## Vogel's Approximation Method

|           | <b>D1</b> | <b>D2</b> | <b>D3</b> | <b>D4</b> | Supply    |
|-----------|-----------|-----------|-----------|-----------|-----------|
| <b>S1</b> | 6         | 1         | 9         | 3         | <b>70</b> |
| <b>S2</b> | 11        | 5         | 2         | 8         | <b>55</b> |
| <b>S3</b> | 10        | 12        | 4         | 7         | <b>70</b> |
| <b>S4</b> | 0         | 0         | 0         | 0         | <b>20</b> |
| Demand    | <b>85</b> | <b>35</b> | <b>50</b> | <b>45</b> |           |



# Unbalanced Transportation problems

|         | D1 | D2 | D3 | D4 | Supply | Penalty |
|---------|----|----|----|----|--------|---------|
| S1      | 6  | 1  | 9  | 3  | 70     |         |
| S2      | 11 | 5  | 2  | 8  | 55     |         |
| S3      | 10 | 12 | 4  | 7  | 70     |         |
| S4      | 0  | 0  | 0  | 0  | 20     |         |
| Demand  | 85 | 35 | 50 | 45 |        |         |
| Penalty |    |    |    |    |        |         |

|         | D1 | D2 | D3 | D4 | Supply | Penalty |
|---------|----|----|----|----|--------|---------|
| S1      | 6  | 1  | 9  | 3  | 70     |         |
| S2      | 11 | 5  | 2  | 8  | 55     |         |
| S3      | 10 | 12 | 4  | 7  | 70     |         |
| Demand  | 65 | 35 | 50 | 45 |        |         |
| Penalty |    |    |    |    |        |         |

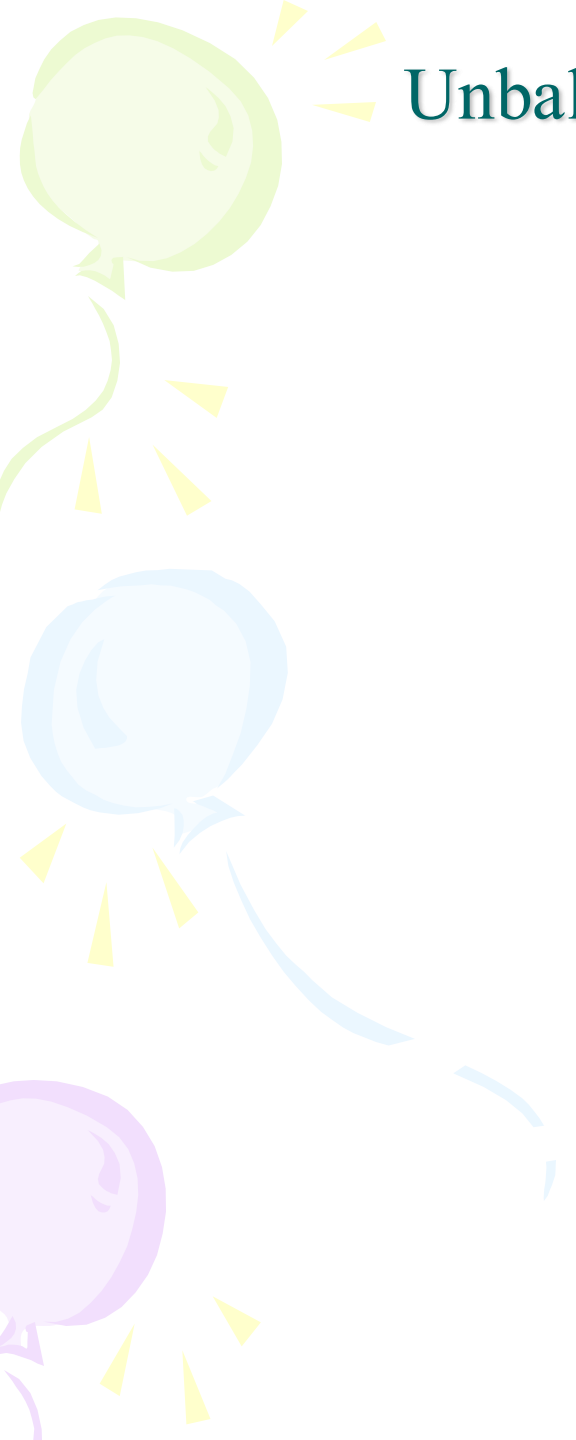


# Unbalanced Transportation problems

|           | <b>D1</b> | <b>D3</b> | <b>D4</b> | Supply    | Penalty |
|-----------|-----------|-----------|-----------|-----------|---------|
| <b>S1</b> | 6         | 9         | 3         | <b>35</b> |         |
| <b>S2</b> | 11        | 2         | 8         | <b>55</b> |         |
| <b>S3</b> | 10        | 4         | 7         | <b>70</b> |         |
| Demand    | <b>65</b> | <b>50</b> | <b>45</b> |           |         |
| Penalty   |           |           |           |           |         |

|           | <b>D1</b> | <b>D4</b> | Supply    | Penalty |
|-----------|-----------|-----------|-----------|---------|
| <b>S1</b> | 6         | 3         | <b>35</b> |         |
| <b>S2</b> | 11        | 8         | <b>5</b>  |         |
| <b>S3</b> | 10        | 7         | <b>70</b> |         |
| Demand    | <b>65</b> | <b>45</b> |           |         |
| Penalty   |           |           |           |         |

|           | <b>D1</b> | <b>D4</b> | Supply    | Penalty |
|-----------|-----------|-----------|-----------|---------|
| <b>S2</b> | 11        | 8         | <b>5</b>  |         |
| <b>S3</b> | 10        | 7         | <b>70</b> |         |
| Demand    | <b>65</b> | <b>10</b> |           |         |
| Penalty   |           |           |           |         |



# Unbalanced Transportation problems

|           | <b>D1</b> | Supply    |
|-----------|-----------|-----------|
| <b>S2</b> | 11        | <b>5</b>  |
| <b>S3</b> | 10        | <b>60</b> |
| Demand    | <b>65</b> |           |



# Unbalanced Transportation problems

Example of unbalanced problem

|           | <b>D1</b> | <b>D2</b> | <b>D3</b> | Supply    |
|-----------|-----------|-----------|-----------|-----------|
| <b>S1</b> | 6         | 1         | 9         | <b>70</b> |
| <b>S2</b> | 11        | 5         | 2         | <b>55</b> |
| <b>S3</b> | 10        | 12        | 4         | <b>70</b> |
| Demand    | <b>85</b> | <b>35</b> | <b>50</b> |           |

$$\sum Supplies = 70 + 55 + 70 = 195$$

$$\sum Demands = 85 + 35 + 50 = 170$$

$$\sum Demands < \sum Supplies$$

$$Excess = \sum Supplies - \sum Demand = 195 - 170 = 25$$

|           | <b>D1</b> | <b>D2</b> | <b>D3</b> | D4        | Supply    |
|-----------|-----------|-----------|-----------|-----------|-----------|
| <b>S1</b> | 6         | 1         | 9         | <b>0</b>  | <b>70</b> |
| <b>S2</b> | 11        | 5         | 2         | <b>0</b>  | <b>55</b> |
| <b>S3</b> | 10        | 12        | 4         | <b>0</b>  | <b>70</b> |
| Demand    | <b>85</b> | <b>35</b> | <b>50</b> | <b>25</b> |           |

Now,  $\sum Demands = \sum Supplies, 195 = 195$



# Unbalanced Transportation problems

NWCM

|        | D1 | D2 | D3 | D4 | Supply |
|--------|----|----|----|----|--------|
| S1     | 6  | 1  | 9  | 0  | 70     |
| S2     | 11 | 5  | 2  | 0  | 55     |
| S3     | 10 | 12 | 4  | 0  | 70     |
| Demand | 85 | 35 | 50 | 25 |        |

|        | D1 | D2 | D3 | D4 | Supply |  |        | D2 | D3 | D4 | Supply |
|--------|----|----|----|----|--------|--|--------|----|----|----|--------|
| S2     | 11 | 5  | 2  | 0  | 55     |  | S2     | 5  | 2  | 0  | 40     |
| S3     | 10 | 12 | 4  | 0  | 70     |  | S3     | 12 | 4  | 0  | 70     |
| Demand | 15 | 35 | 50 | 25 |        |  | Demand | 35 | 50 | 25 |        |



# Unbalanced Transportation problems

|  |        |    |    |        |  |  |        |    |    |        |  |
|--|--------|----|----|--------|--|--|--------|----|----|--------|--|
|  |        | D3 | D4 | Supply |  |  |        | D3 | D4 | Supply |  |
|  | S2     | 2  | 0  | 5      |  |  | S3     | 4  | 0  | 70     |  |
|  | S3     | 4  | 0  | 70     |  |  | Demand | 45 | 25 |        |  |
|  | Demand | 50 | 25 |        |  |  |        |    |    |        |  |